

Atmosphere TV

DOOH Venue Incrementality & Media-Mix Measurement

A causal measurement framework, calibrated media-mix model, budget optimizer, and venue-revenue model — built to answer the questions Atmosphere's advertising business and its own network economics run on.



Interview technical demo • Senior Data Scientist

Three questions this project answers

1

Did the campaign actually work?

Isolate the TRUE incremental foot traffic caused by ad exposure — net of seasonality, trend, and each venue's own baseline pattern.

The sell-side differentiator: measurement Atmosphere's go-to-market team can take to market and clients can trust.

2

How should budget be spent?

Given a fixed weekly budget, how should it split across restaurants, gyms, bars, and waiting rooms — accounting for each venue type's own diminishing-returns curve?

The media-planning / pricing question advertisers ask before they commit spend.

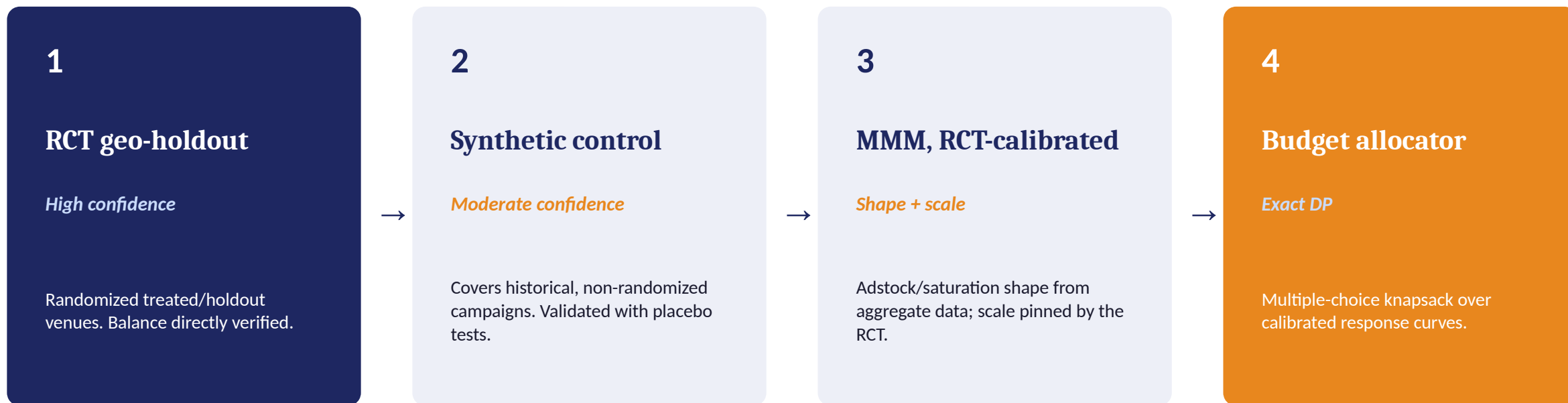
3

Where's Atmosphere's own revenue upside?

Which existing venues are under-monetized relative to their own traffic and quality — and which prospective venues are worth prioritizing for network expansion?

Atmosphere's own buy-side question, not the advertiser's.

Design: one pipeline, confidence flows from experiment to product



Why this shape: a single trusted experimental read (RCT) is expensive to run everywhere, so it's used to calibrate a richer but less-trusted model (MMM) that covers the whole network — and that calibrated model then powers a budget decision.

Synthetic data with a known, injected ground truth

4 venue types × 100 venues each, 104 weeks (52 pre-period + 52 campaign)

RCT pool

Randomly split treated / holdout within venue_type × geo_cluster × traffic-tier strata for a fixed benchmark campaign. This is a real Atmosphere-designed measurement product, not a historical campaign.

Used to anchor confidence — assignment is verified, not assumed.

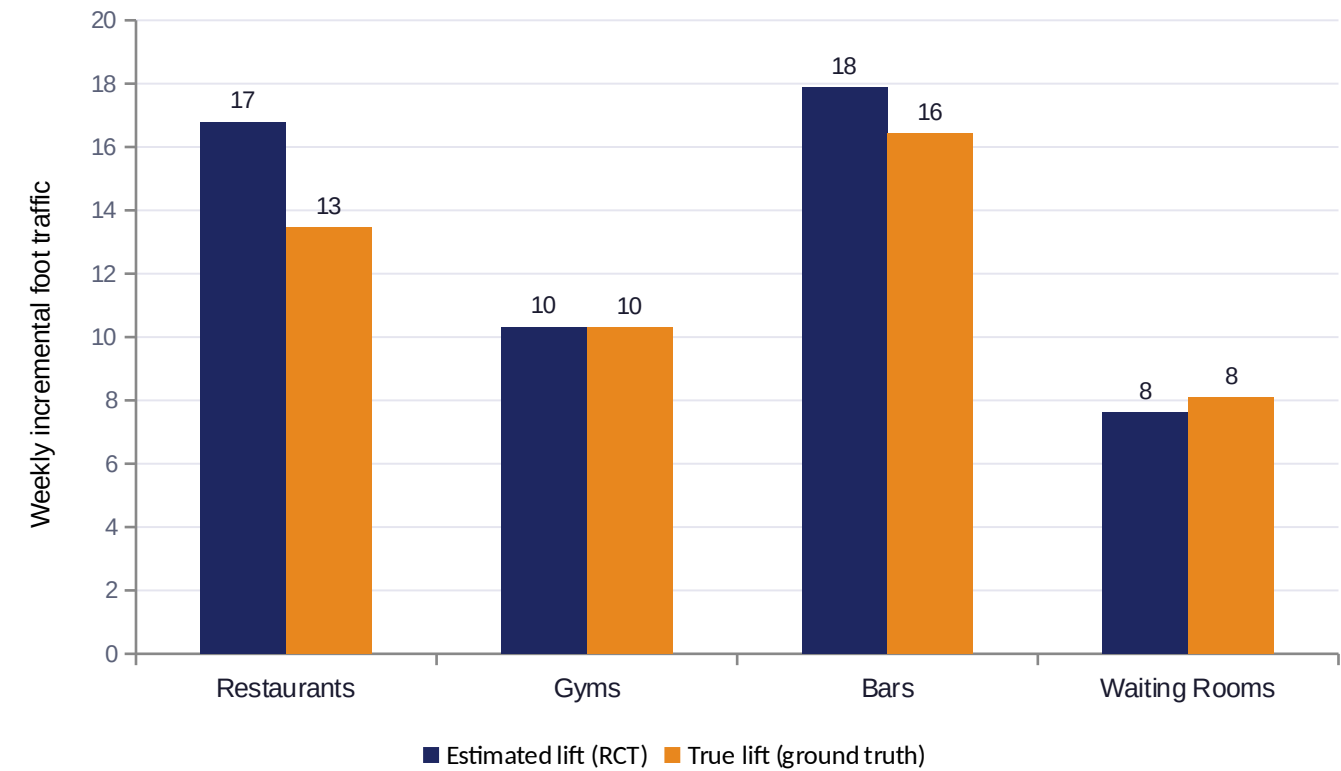
Observational pool

Advertisers activate venues themselves — non-randomly. Higher-baseline-traffic venues are systematically more likely to be activated (a real selection mechanism), and weekly frequency varies venue-to-venue.

A naive before/after comparison here is confounded by selection — this is what synthetic control and the MMM have to work around.

RCT geo-holdout — high confidence

2/16 balance tests sig.

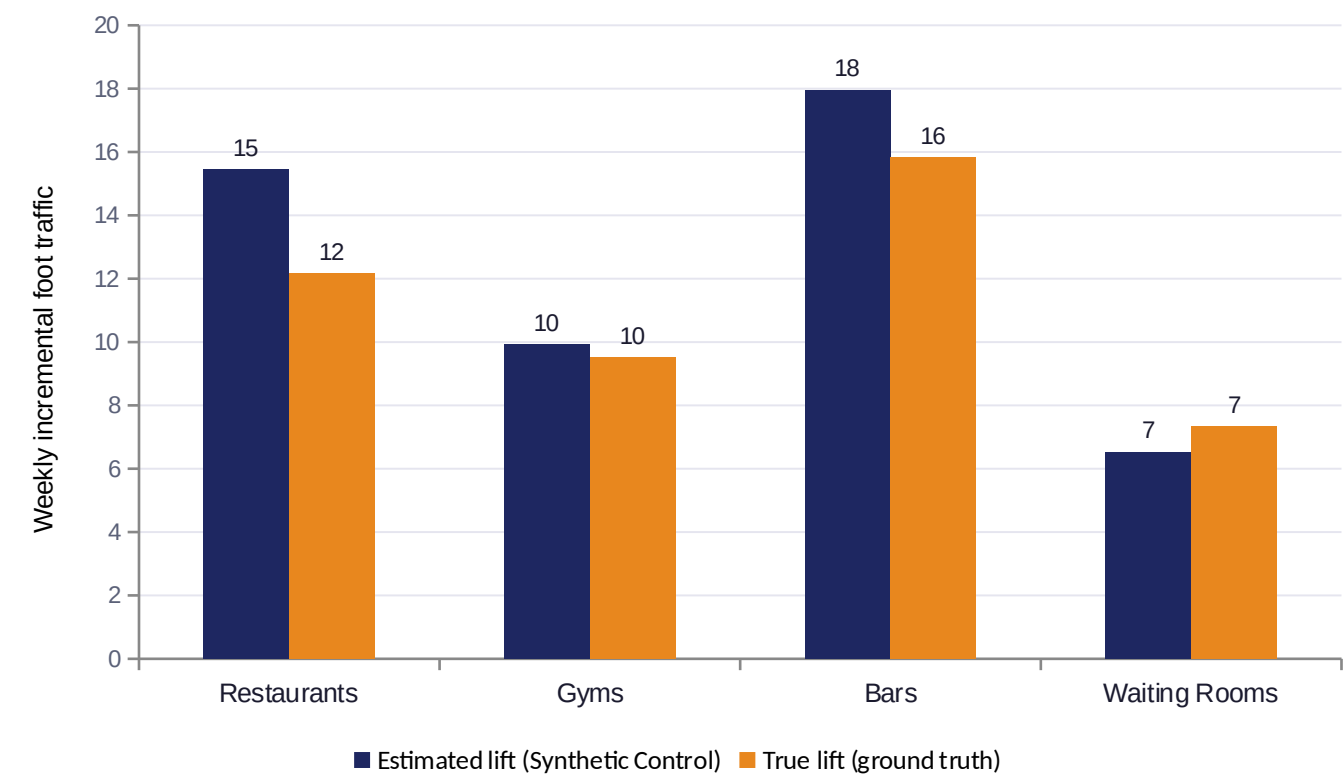


Venue type	95% CI	p-value
Restaurants	[11.1, 22.4]	< 0.001
Gyms	[5.2, 15.4]	< 0.001
Bars	[13.3, 22.5]	< 0.001
Waiting Rooms	[5.1, 10.2]	< 0.001

Pre-treatment balance check: only 2 of 16 covariate × venue-type tests were significant at $p \leq 0.05$ — in line with what chance alone predicts. Randomization worked as designed, not assumed.

Synthetic control — moderate confidence

Covers historical, non-randomized campaigns — the identifying assumption can't be directly tested the way randomization can, only checked indirectly.

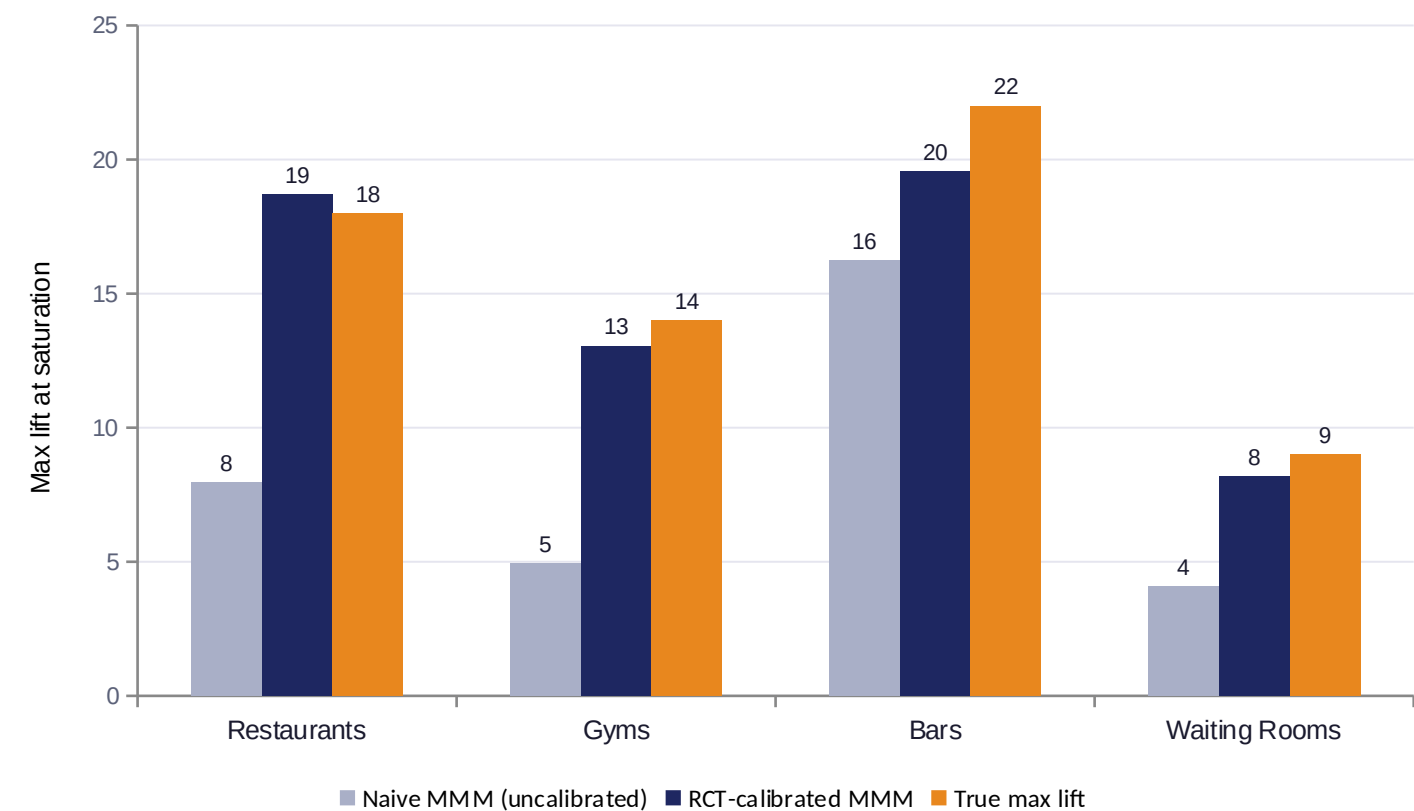


Venue type	Placebo p	Pre-RMSPE
Restaurants	0.160	26.4
Gyms	0.050	14.4
Bars	0.100	17.8
Waiting Rooms	0.110	8.3

Two validation checks, since synthetic control has no closed-form standard error: pre-period fit quality (RMSPE — poor fits are flagged and excluded from the aggregate estimate) and an in-space placebo test on every donor venue.

MMM: an uncalibrated model would have underpriced results

Adstock + saturation shape comes from the aggregate weekly series; scale is pinned by the RCT's high-confidence estimate.



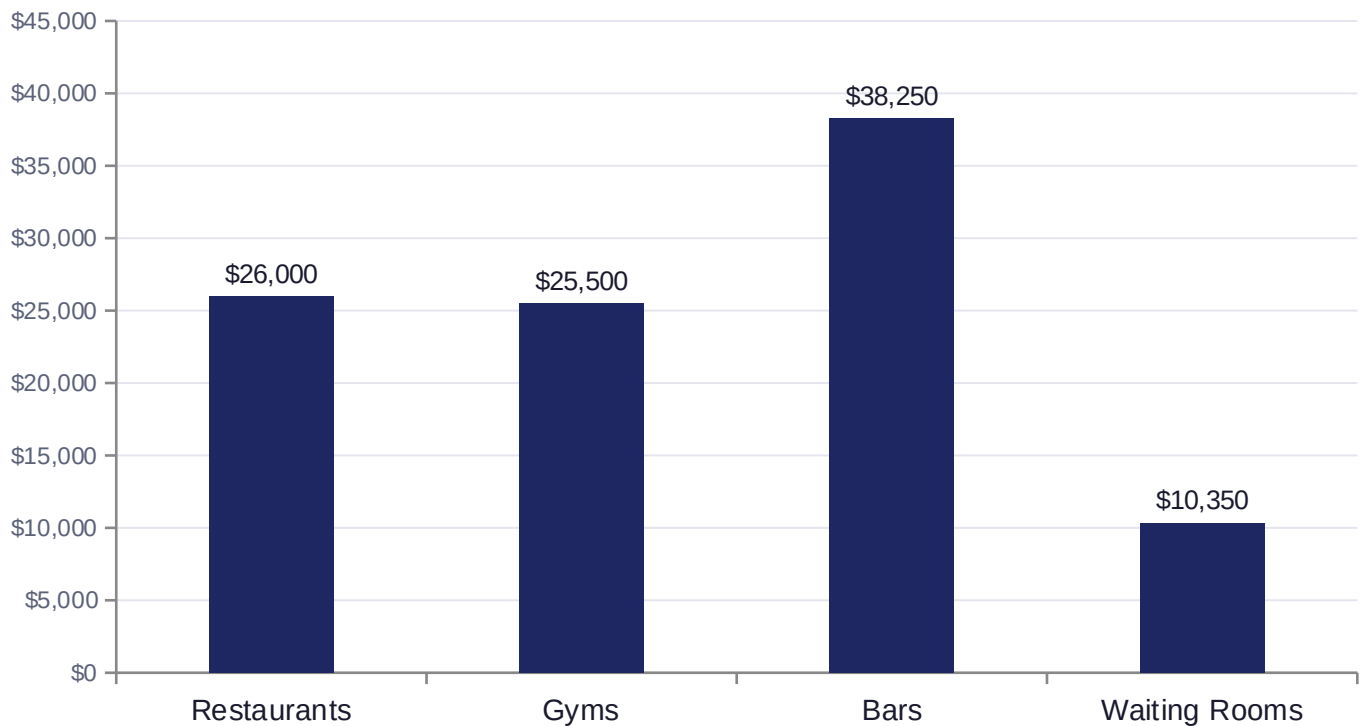
Calibration adjustment

Restaurants	+134%
Gyms	+165%
Bars	+20%
Waiting Rooms	+100%

For 3 of 4 venue types, the naive observational fit understated true incremental lift by 100–165% — the kind of gap real MMM practice cites as its core identification problem, and exactly why the RCT anchor matters.

Budget allocator: exact DP, not a greedy walk

A Hill/S-shaped response curve is CONVEX before its inflection point — a greedy “spend the next dollar on whichever venue type currently looks best” heuristic isn't guaranteed optimal there. An earlier greedy version of this allocator actually underperformed a naive equal-split baseline; the DP formulation below has no concavity requirement and is guaranteed to find the grid-optimal allocation.



\$100,000 weekly budget example

\$20,000 budget **+48.6%**
vs. naive equal-split

\$50,000 budget **+11.1%**
vs. naive equal-split

\$100,000 budget **+2.4%**
vs. naive equal-split

The optimizer's edge is largest when budget is scarce — exactly when allocation decisions matter most.

The other side of the business

One system answers the advertising side. The same system answers the venue-network side too.

Advertising incrementality

- Prove and price advertising incrementality
- Feeds go-to-market: sell-side differentiator, client trust
- RCT + synthetic control + calibrated MMM + budget DP

Feeds the venue-economics model as a validated causal-value input (calibrated per-exposure lift), not a separate silo.

Venue economics

- Predict a venue's realized ad-revenue from its characteristics
- Flag under-monetized existing venues for sales/ops follow-up
- Rank prospective venues for expansion priority
- Gradient-boosted trees, honest out-of-fold evaluation

“Smarter on both sides of the business” — the venue-side model shares its data foundation and causal-value input with the advertising side, rather than being a disconnected second project. Results on the next slide.

Predicting venue ad-revenue, honestly

Gradient-boosted trees on observable venue characteristics + the calibrated per-exposure lift as a feature.

0.84

Held-out R²

22.1%

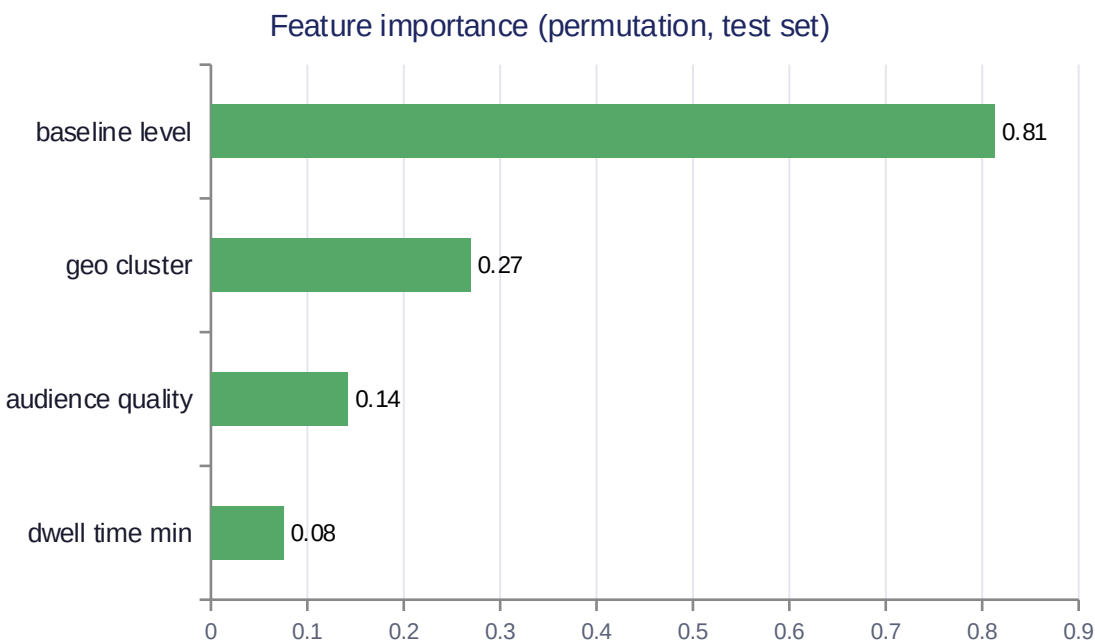
Held-out MAPE

0.94

Corr. w/ latent true potential

90% vs 28%

Flagged-tail latent gap rate



Top under-monetized venues (of 20 flagged)

Venue #85 (Restaurants)	-62%
Venue #309 (Waiting Rooms)	-61%
Venue #53 (Restaurants)	-61%
Venue #247 (Bars)	-60%

Top prospect venues (expansion priority)

P0018 — Bars (existing market)	\$684
P0055 — Restaurants (existing market)	\$524
P0000 — Bars (new market)	\$500
P0001 — Bars (new market)	\$496

Flags come from 5-fold out-of-fold predictions, never a model scoring the venue it was trained on. The calibrated-lift feature carries near-zero importance — honest, since it's constant within venue_type once venue_type itself is a feature.

Honest scope

Synthetic data

Every figure in this project is synthetic, with a known injected ground-truth effect used specifically to validate that each method recovers it before trusting it conceptually. Not a claim about any real company's data.

Media-effectiveness & rate-card figures

Dwell time, screen count, ad rate card, and similar parameters shown are illustrative demo values, not researched real industry benchmarks. In production: Nielsen OOH, DSP data (e.g. Vistar), or Atmosphere's own play logs and rate card.

Multi-touch attribution — deliberately not built

Atmosphere's ambient-screen model has no individual-level, cross-venue touchpoint log by default. Building MTA would require purchased mobile location/device-matching data — a real but non-default assumption, so it's scoped out rather than forced.

Cost & revenue assumptions

The \$/frequency-unit figures behind the budget allocator, and the ad-rate-card behind the venue-revenue model, are illustrative, editable placeholders — in production these come from Atmosphere's own rate card by venue type and daypart.

Takeaways

- 1 Match the method to the assignment mechanism — RCT where randomization is designed, synthetic control where it isn't, each with its own validation check.
- 2 An uncalibrated MMM understated true incremental value by up to 165% here — experiment calibration isn't optional polish, it's the identification fix.
- 3 Optimization needs the right algorithm for the curve's shape — a greedy heuristic silently lost to a naive baseline on non-concave response curves; the exact DP formulation doesn't.
- 4 One connected system, not two projects: the venue-revenue model reuses the causal pipeline's per-exposure value as an input, recovers latent ground truth at 0.95 correlation, and cleanly separates market-level effects from venue-level execution gaps.

Thank you